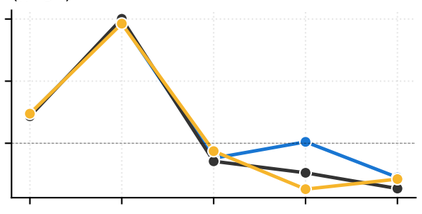
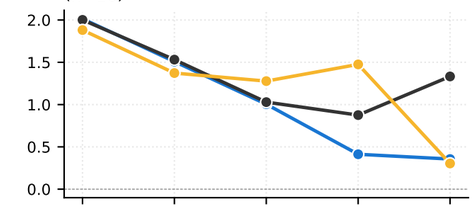
Supplementary Figure SB′ — trajectories for every qualifying subsystem

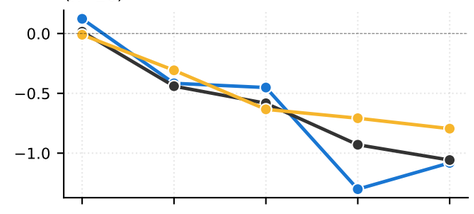
Acyl-CoA hydrolysis (n=54)



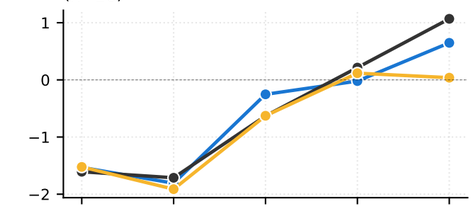
Acylglycerides metabolism (n=16)



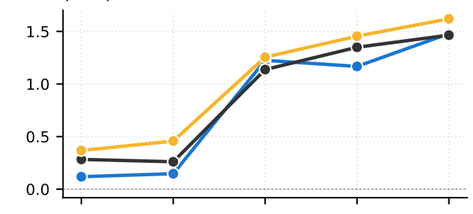
Aminoacyl-tRNA biosynthesis (n=20)



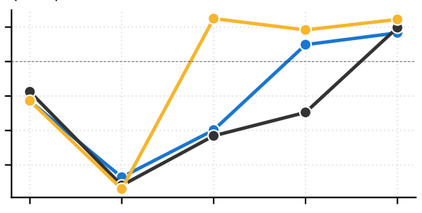
Arachidonic acid metabolism (n=28)



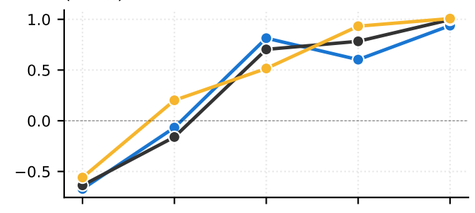
Beta oxidation of di-unsaturated fatty acids (n-6) (mitochondrial) (n=6)



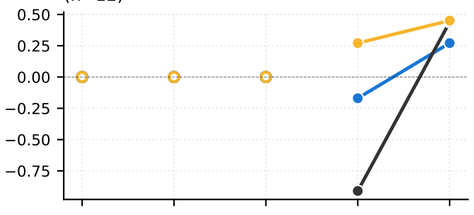
Beta oxidation of di-unsaturated fatty acids (n-6) (peroxisomal) (n=6)



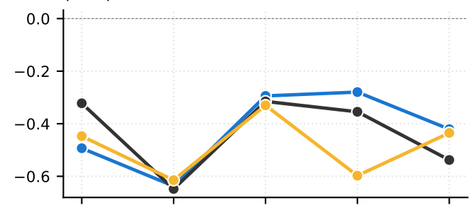
Beta oxidation of even-chain fatty acids (mitochondrial) (n=23)



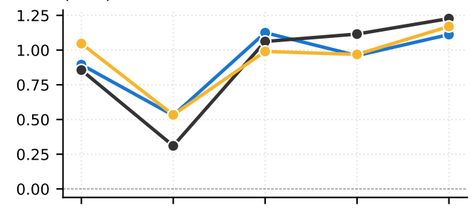
Beta oxidation of even-chain fatty acids (peroxisomal) (n=12)



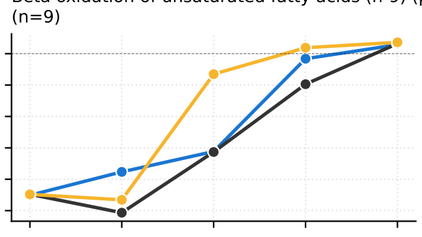
Beta oxidation of odd-chain fatty acids (mitochondrial) (n=8)



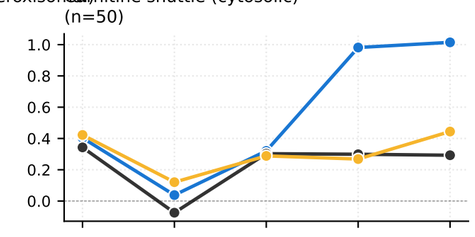
Beta oxidation of unsaturated fatty acids (n-9) (mitochondrial) (n=5)



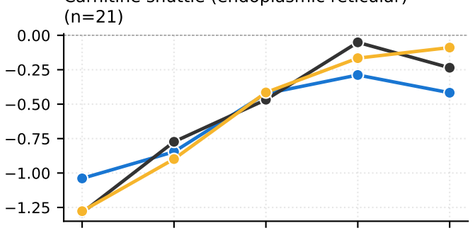
Beta oxidation of unsaturated fatty acids (n-9) (peroxisomal) (n=9)



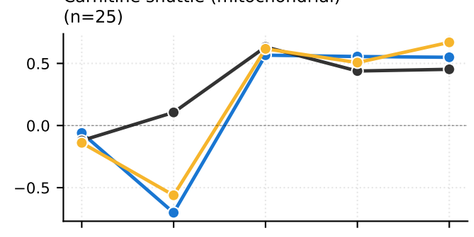
Carnitine shuttle (cytosolic) (n=50)



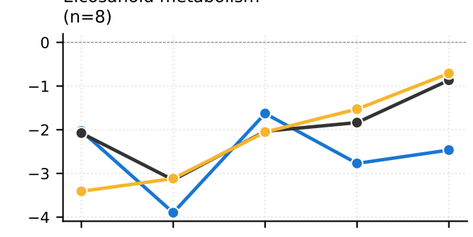
Carnitine shuttle (endoplasmic reticular) (n=21)



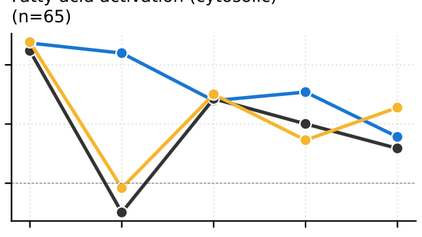
Carnitine shuttle (mitochondrial) (n=25)



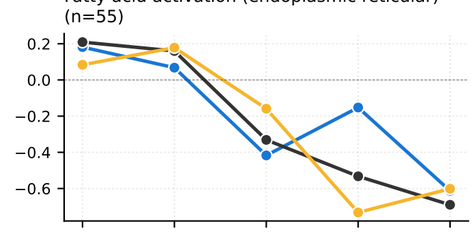
Eicosanoid metabolism (n=8)



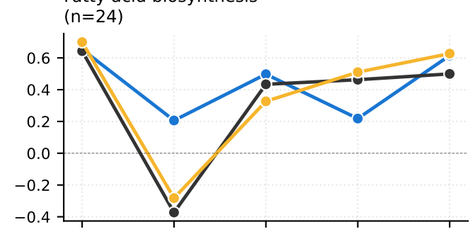
Fatty acid activation (cytosolic) (n=65)



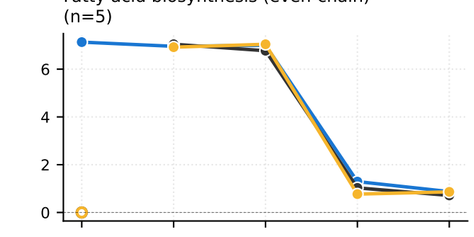
Fatty acid activation (endoplasmic reticular) (n=55)



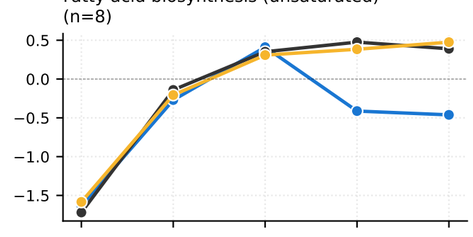
Fatty acid biosynthesis (n=24)



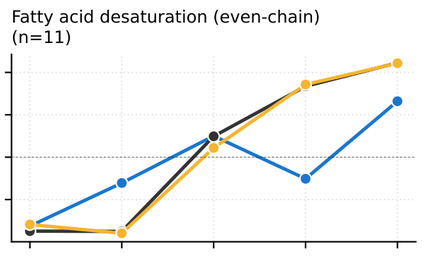
Fatty acid biosynthesis (even-chain) (n=5)



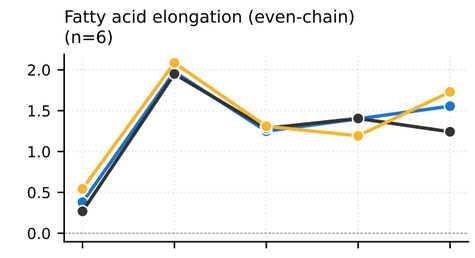
Fatty acid biosynthesis (unsaturated) (n=8)



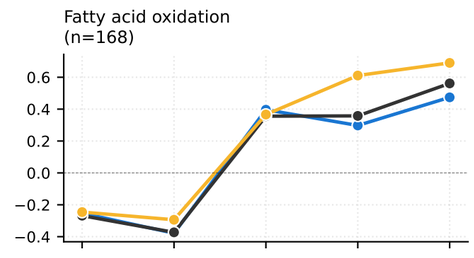
Fatty acid desaturation (even-chain) (n=11)



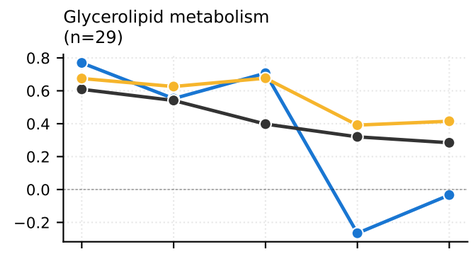
Fatty acid elongation (even-chain) (n=6)



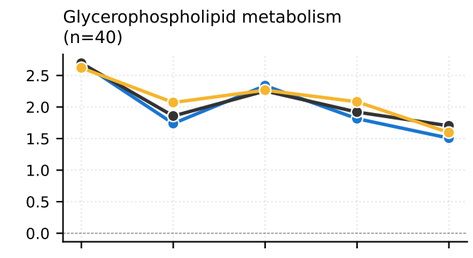
Fatty acid oxidation (n=168)



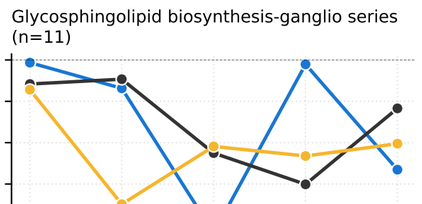
Glycerolipid metabolism (n=29)



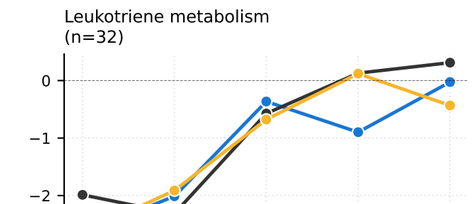
Glycerophospholipid metabolism (n=40)



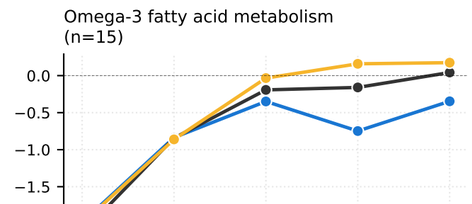
Glycosphingolipid biosynthesis-ganglio series (n=11)



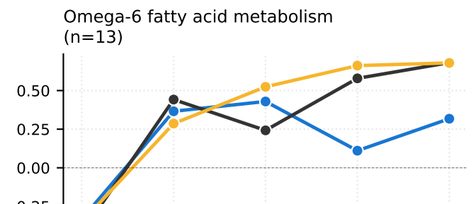
Leukotriene metabolism (n=32)



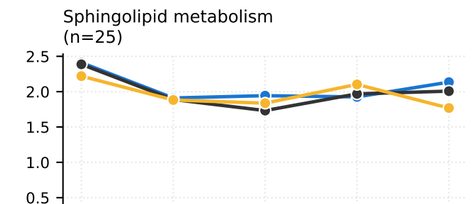
Omega-3 fatty acid metabolism (n=15)



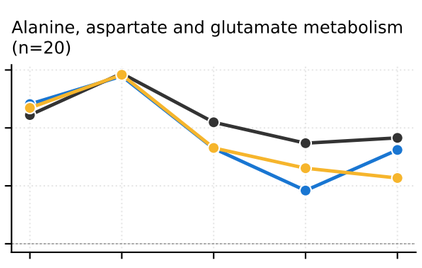
Omega-6 fatty acid metabolism (n=13)



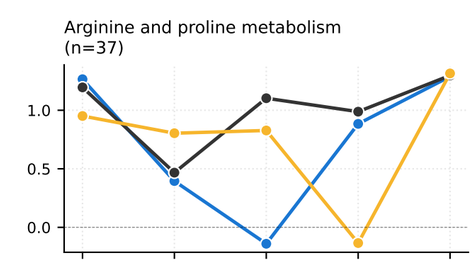
Sphingolipid metabolism (n=25)



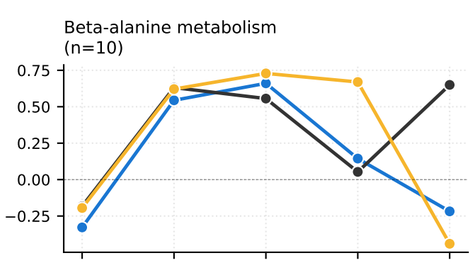
Alanine, aspartate and glutamate metabolism (n=20)



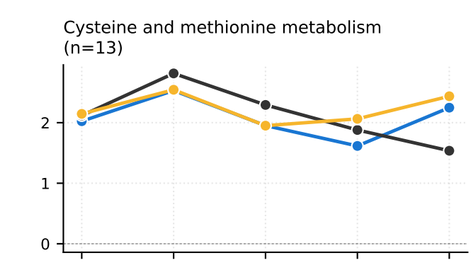
Arginine and proline metabolism (n=37)



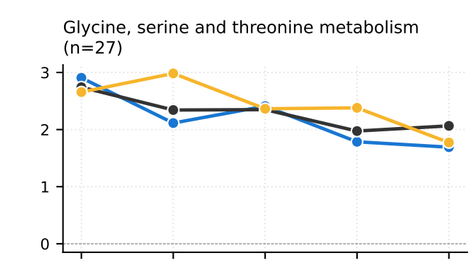
Beta-alanine metabolism (n=10)



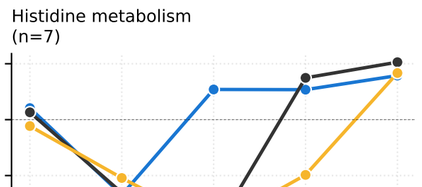
Cysteine and methionine metabolism (n=13)



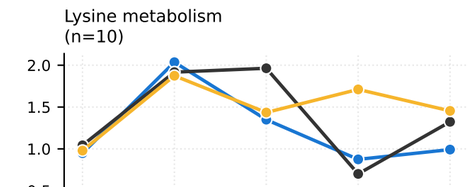
Glycine, serine and threonine metabolism (n=27)



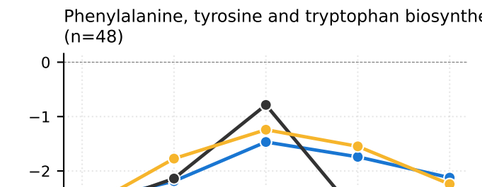
Histidine metabolism (n=7)



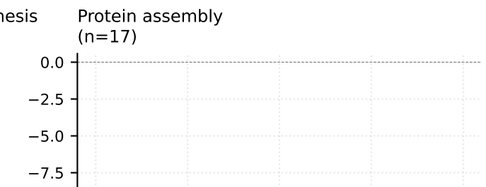
Lysine metabolism (n=10)



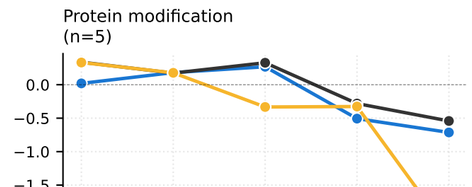
Phenylalanine, tyrosine and tryptophan biosynthesis (n=48)



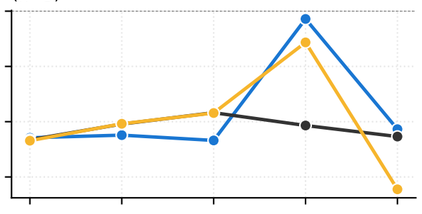
Protein assembly (n=17)



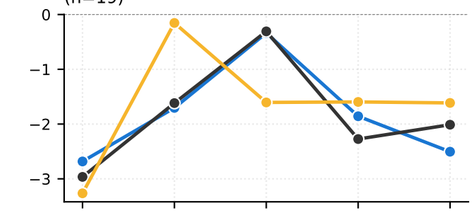
Protein modification (n=5)



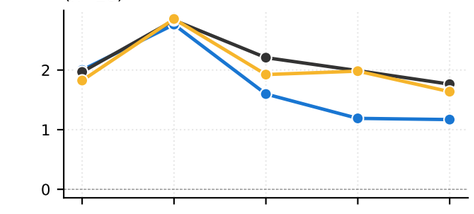
Tryptophan metabolism (n=5)



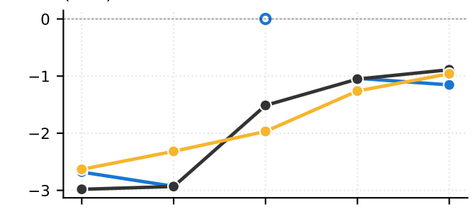
Tyrosine metabolism (n=19)



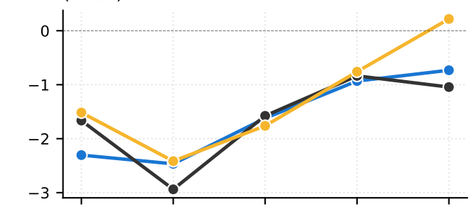
Valine, leucine, and isoleucine metabolism (n=28)



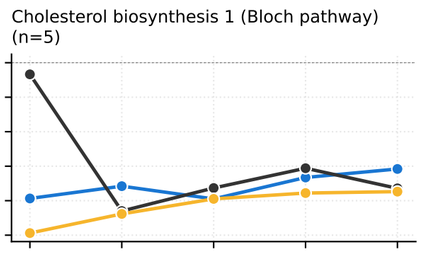
Androgen metabolism (n=7)



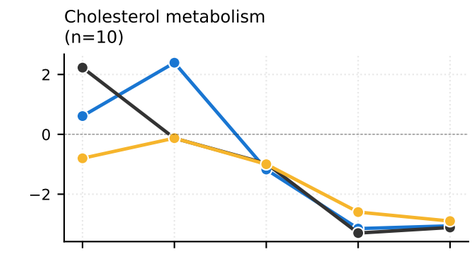
Bile acid biosynthesis (n=26)



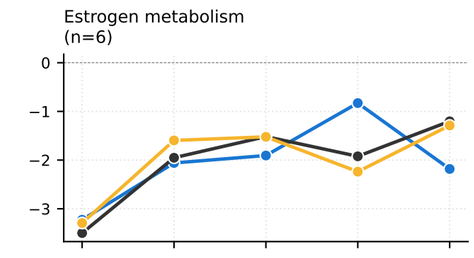
Cholesterol biosynthesis 1 (Bloch pathway) (n=5)



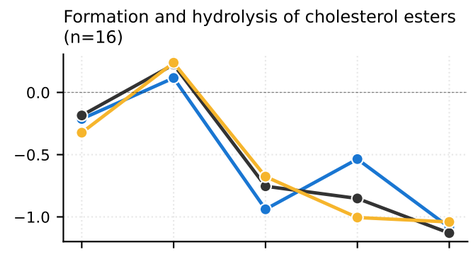
Cholesterol metabolism (n=10)



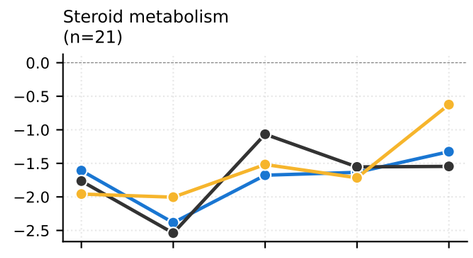
Estrogen metabolism (n=6)



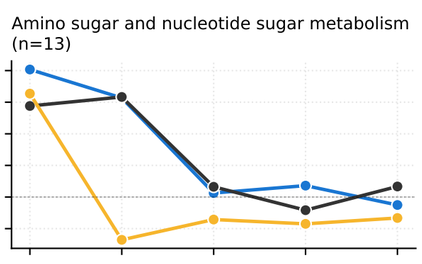
Formation and hydrolysis of cholesterol esters (n=16)



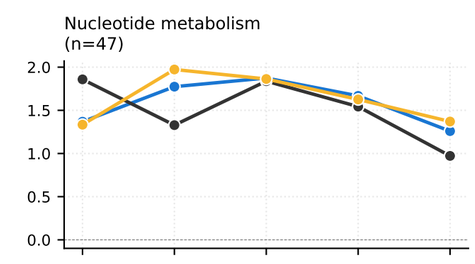
Steroid metabolism (n=21)



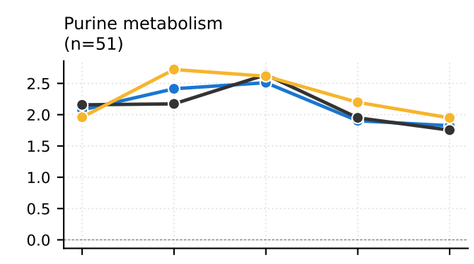
Amino sugar and nucleotide sugar metabolism (n=13)



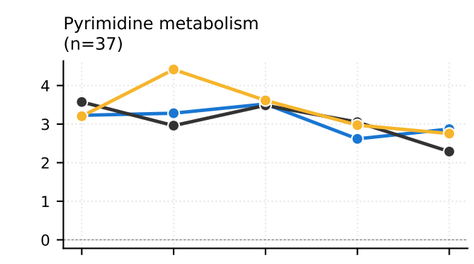
Nucleotide metabolism (n=47)



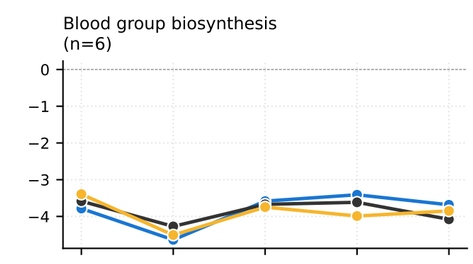
Purine metabolism (n=51)



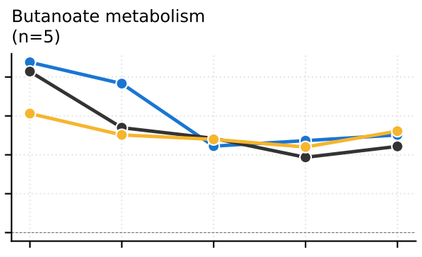
Pyrimidine metabolism (n=37)



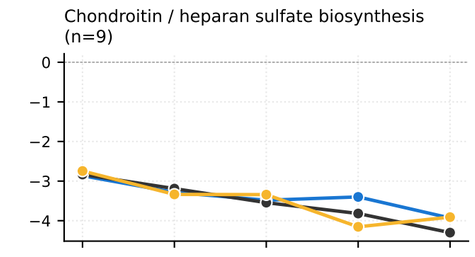
Blood group biosynthesis (n=6)



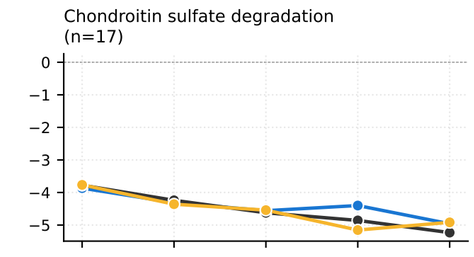
Butanoate metabolism (n=5)



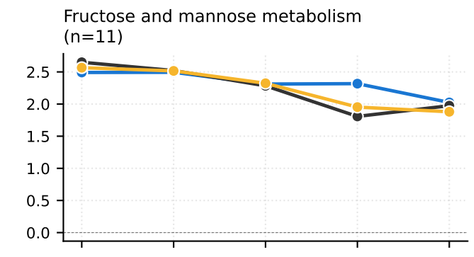
Chondroitin / heparan sulfate biosynthesis (n=9)



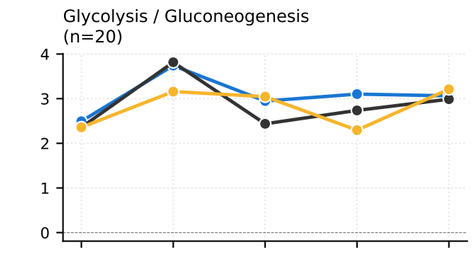
Chondroitin sulfate degradation (n=17)



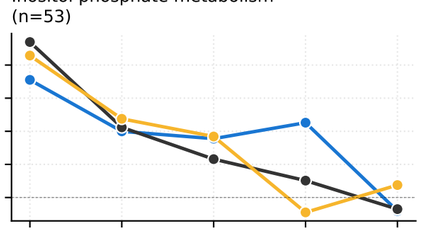
Fructose and mannose metabolism (n=11)



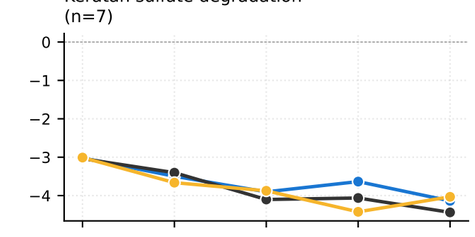
Glycolysis / Gluconeogenesis (n=20)



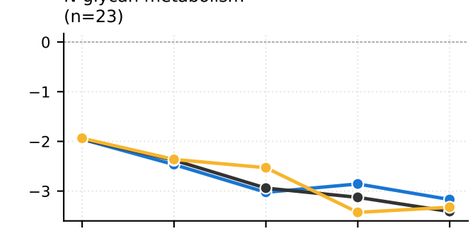
Inositol phosphate metabolism (n=53)



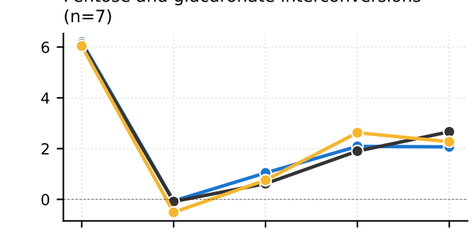
Keratan sulfate degradation (n=7)



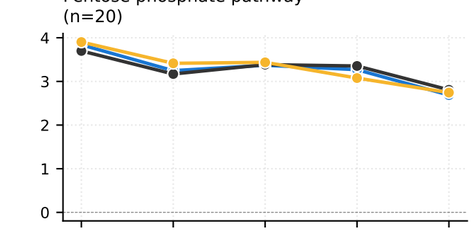
N-glycan metabolism (n=23)



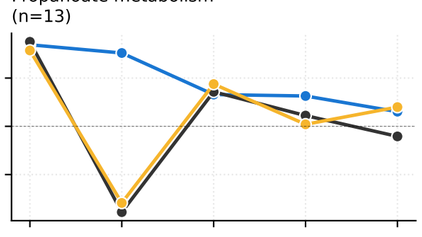
Pentose and glucuronate interconversions (n=7)



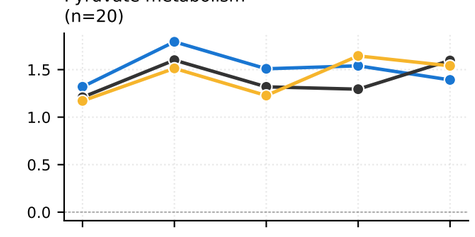
Pentose phosphate pathway (n=20)



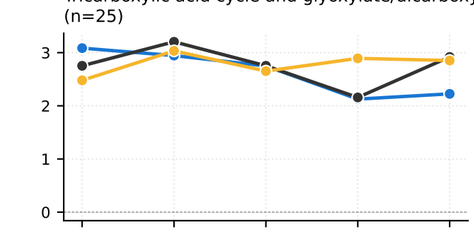
Propanoate metabolism (n=13)



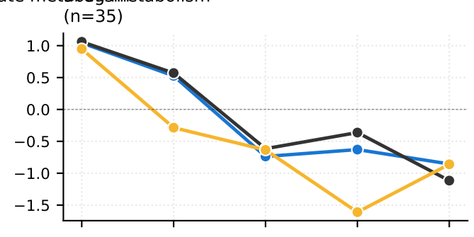
Pyruvate metabolism (n=20)



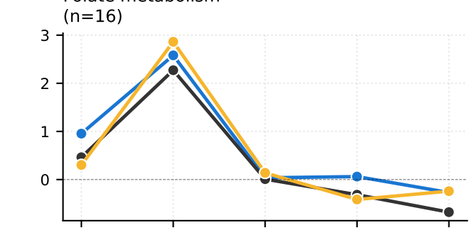
Tricarboxylic acid cycle and glyoxylate/dicarboxylate metabolism (n=25)



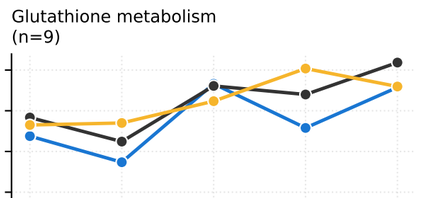
Urea cycle metabolism (n=35)



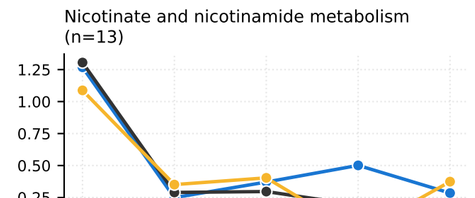
Folate metabolism (n=16)



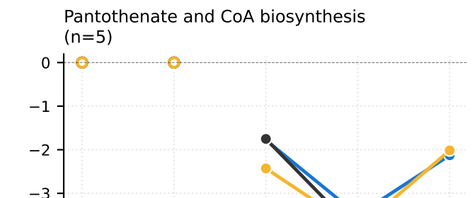
Glutathione metabolism (n=9)



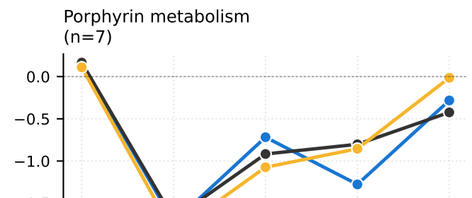
Nicotinate and nicotinamide metabolism (n=13)



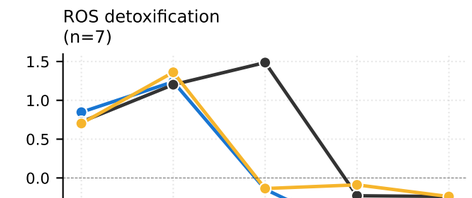
Pantothenate and CoA biosynthesis (n=5)



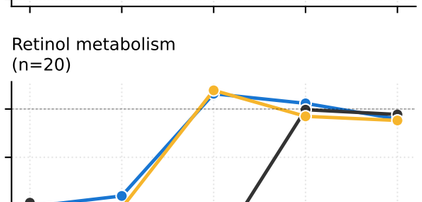
Porphyrin metabolism (n=7)



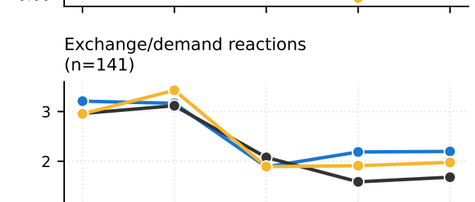
ROS detoxification (n=7)



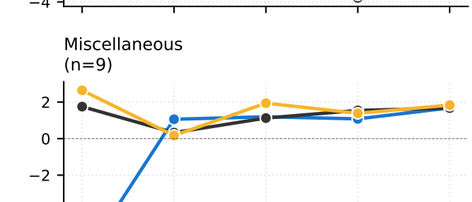
Retinol metabolism (n=20)



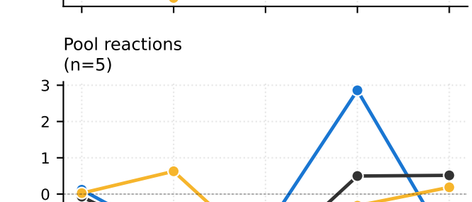
Exchange/demand reactions (n=141)



Miscellaneous (n=9)



Pool reactions (n=5)



Transport reactions (n=739)

